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Semantic Model-based Systems Engineering based on KARMA: A Research and Practice Roadmap 2025

Jinzhi Lu

École Polytechnique Fédérale de Lausanne
Jinzhi.lu@epfl.ch

Yan Yan

Beijing Institute of Technology

Feng Lei

KTH Royal Institute of Technology

Xijin Tang

CAS Academy of Mathematics and Systems
Science

Junjie Tang

China Aerospace Science and Technology
Corporation

Huisheng Zhang

Shanghai Jiaotong University

Martin Törngren

KTH Royal Institute of Technology

Yves Keraron

Innovation Strategy and Digital Engineering
Using Semantics

Dimitris Kiritsis

École Polytechnique Fédérale de Lausanne

Guoxin Wang

Beijing Institute of Technology

David Cameron

University of Oslo

Yangyang Zhang

China Electronics Standardization Institute

Jian Wang

University of Electronic Science and
Technology of China

Jian Tang

COMAC BATRI

Michel Reniers

Eindhoven University of Technology

Junda Ma

Beijing Institute of Technology

Shouxuan Wu

Beijing Institute of Technology

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Abstract. Model-based Systems Engineering is proposed as a graphical approach to support formalization of system artifacts across system lifecycle based on models since 1993. The previous motivation of graphical specification is to provide a unified graphical description on the perspective of systems engineering in order to formalize system architectural views, to promote communications among stakeholders and to support system analysis and verification. However, when different modeling tools are developed based on such graphical specifications, model and data interoperability across modeling tools is the biggest challenge faced by the tool vendors and MBSE practitioners. Thus, semantic specification is proposed again to enhance data interoperability, such as SysML 2.0 as a possible solution. In this paper, we propose a new semantic MBSE language and framework aiming to support complex system development using a semantic two-core mechanism: 1) KARMA language and 2) Industrial Ontologies Foundry (IOF) SE and MBSE ontology. Then, we introduce the KARMA Roadmap 2025 including technical vision, organizational views and standardization. A

new KARMA open-source modeling environment is planned to be created in order to provide MBSE application for more industrial practitioners.

Introduction

Model-based systems engineering (MBSE) is considered as a next generation systems engineering technology to support complex system development using models. Using MBSE, system architecture, system lifecycle and system artifacts are represented using graphical representations in a unified specification (Micouin 2014). Initially, the graphical specifications were proposed to provide a unified description across different domains in order to replace documents and codes. However, since the MBSE is widely used, data interoperability is becoming a big challenge when integrating modeling languages, modeling tools and modeling methodologies (Lu et al. 2021). Thus, modeling approaches based on semantic contents, ontology for MBSE and semantic modeling are widely proposed to realize the integrated modeling for the entire lifecycle of complex system development.

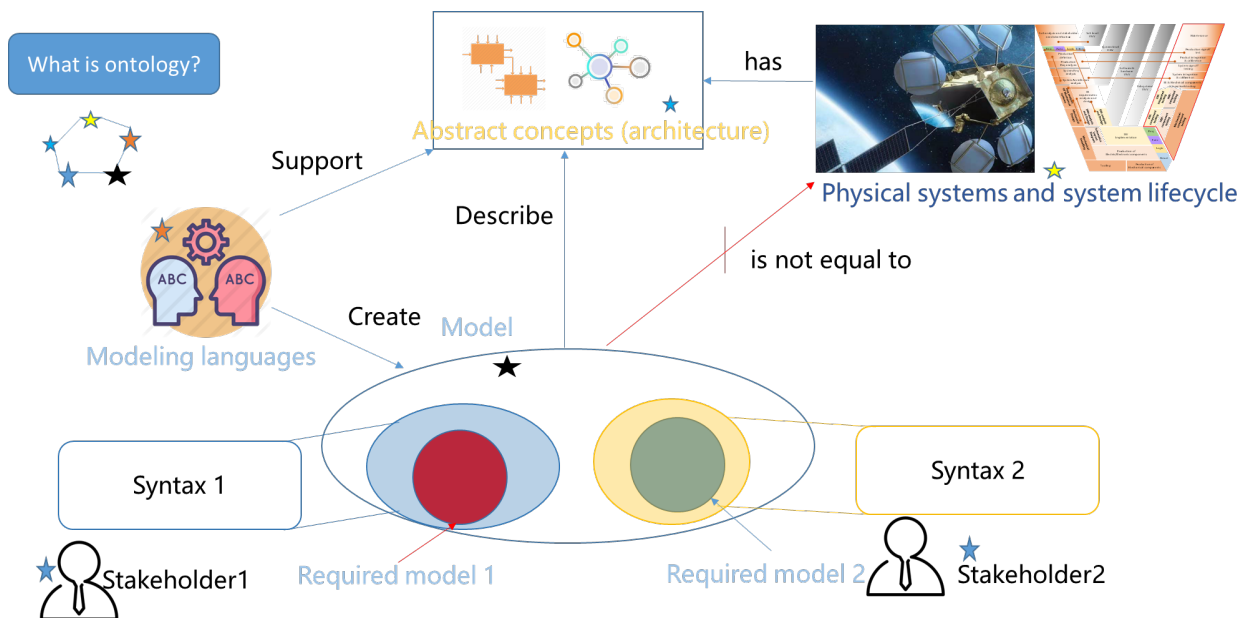


Figure 1 Ontology, systems engineering and models

Ontology refers to a representational artifact, including a taxonomy as each proper part, whose representations are intended to designate some combination of system lifecycle, system artifact, stakeholders, modeling languages, and virtual models for systems, defined classes, and certain relations between them (Arp, Smith, and Spear 2016). As the figure shows, the aim of ontology is to represent entities of virtual models, and their real physical systems or development processes (Mario Augusto Bunge 1980). Ontology (semantic) models refer to a data structure, mathematical or graphical descriptions for representing entities and the topology between them using the ontology which are developed based on modeling languages and specifications. Systems engineering formalism (van Beek et al. 2006) and MBSE formalism (Lu et al. 2021) are defined as a formal means to specify the system artifacts, system lifecycle and virtual MBSE models.

In this paper, we propose a semantic MBSE research and practice roadmap 2025 based on two core mechanisms: KARMA (Kombination of ARchitecture Model specificAtion) language and Industrial Ontologies Foundry (IoF) ¹ SE and MBSE ontology. KARMA language is a semantic modeling language for a multi-architecture modeling approach (Lu, Wang, et al. 2020). It supports to formalize multiple system architecture languages, such as SysML, etc. Moreover, KARMA is expected to support descriptions and simulation of different architectural views of systems engineering. The IoF

¹ Industrial Ontologies Foundry (IoF) Systems Engineering working group <https://www.industrialontologies.org/systems-engineering-wg/>

SE and MBSE ontology refers to the ontology supporting MBSE and systems engineering which is developed by IoF systems engineering working group. This roadmap includes three parts: 1) Technical visions towards to full lifecycle modeling based on the two core specifications; 2) Organizational support for KARMA use; 3) standardized vision for data interoperability. The rest of this paper is organized as follows. We demonstrate our related work in Section II. Then the roadmap of semantic MBSE content based on KARMA language is represented in Section III. Finally, gaps and limitations are investigated and future work is demonstrated in Section IV and we offer the conclusion in Section V.

Related Work

In this section, we investigate the ontology mentioned in the last 5-year’s MBSE references. Moreover, we also have a brief state of the art about the semantic modeling related to important MBSE languages and methodologies.

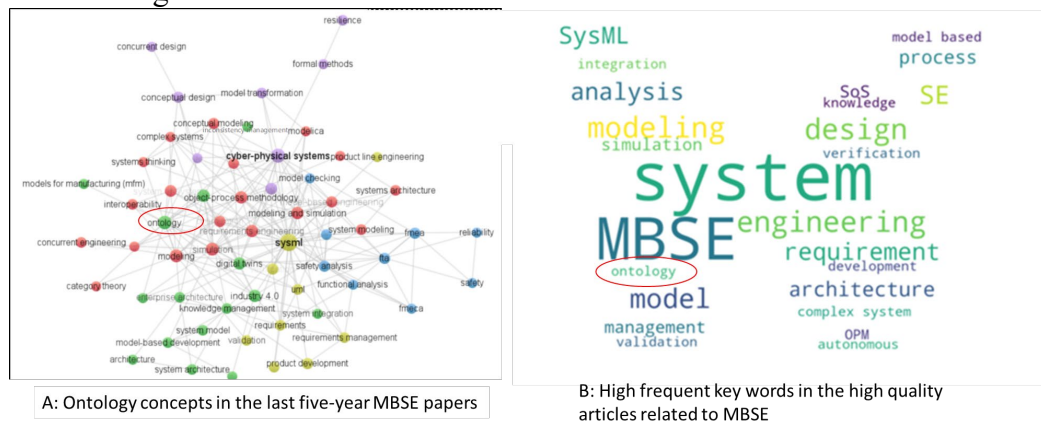


Figure 2 Ontology mentioned in the last 5-year MBSE related papers

Is semantic modeling important? Our previous research articles have several surveys on research articles on MBSE. As shown in Figure 2, among 1074 papers (searched by key words "Model-based Systems Engineering" in Scopus database and INCOSE international symposium from 2015), ontology takes an important role in the MBSE domain (Li et al. 2021). Particularly, compared with the MBSE articles since 1993, ontology is becoming more important recently. Moreover, in another survey about the high quality articles of MBSE in IEEE and INCOSE communities, ontology is one of the most highly frequent key words existing in the almost 150 articles (Mengru Dong, Lu Jinzhi, Guoxin Wang, Xiaochen Zheng 2022). From these two surveys, we find ontology is becoming as important as semantic modeling, since the last five years. Most of these papers focus on data interoperability when using MBSE.

What is semantic MBSE? In this paper, semantic MBSE refers to an approach making use of the MBSE language which is based on a semantic specification or developed semantic content to construct MBSE models rather than graphical specifications for notation development of MBSE models. Through such semantic specifications, different meta-models can be created in order to support MBSE model development across the entire lifecycle. Moreover, MBSE models can also be described by such semantic specifications. Compared with MBSE based on the graphical specifications, semantic MBSE has several benefits:

- Improve precision and expressiveness of the MBSE language concepts
- Support consistency and integration among different language concepts and tools
- Models and tools interoperability

State of the art. Currently, semantic modeling and modeling approach based on semantic specifications are widely provided to promote data interoperability. Semantic modeling is a modeling technique used to depict the relationships that exist among specific values of data, such as the data

modeling diagram, but now focused on the values of the data and the relationship they have (Luisi 2014). Modeling based on semantic specifications is a means to provide a semantic specification to formalize the model structure and topology such as SysML 2.0 (Gray and Rumpe 2021). We encountered several MBSE modeling languages which are used to support architecture formalism, physical structure formalism and process formalism with their developed semantic specifications. As shown in Table 1, modeling languages, methodologies and their semantic specifications are listed.

Table 1: MBSE languages and their semantic specifications.

MBSE Language	MBSE methodology	Semantic specification
SysML 1.x (Friedenthal, Moore, and Steiner 2015)	MagicGrid, IBM Harmony, etc	XMI, RDF
Modelica (Fritzson and Engelson 1998)	Simulation and verification	Modelica specification
Matlab language (MATLAB n.d.)	Simulation and verification	Matlab language specification
CyPhyML and Formula (Sztipanovits et al. 2014)	Architecture design and model integration	CyPhyML and Formula specification
OPL (Blekhman, Wachs, and Dori 2015)	Object process methodology	OPL specification
Capella DSML (Roques et al. 2016)	Arcadia	Semantic specification in textual editor
BPMN (Geiger et al. 2015)	-	XML specification for BPMN
SysML 2.0x (Gray and Rumpe 2021)	-	SysML 2.0x specification

Findings. In summary, semantic modeling, ontology and semantic specifications for different MBSE languages and methodologies are expected to overcome the challenges to promote the data interoperability and accuracy of system description across different models. Except for these two core benefits, other general potential benefits are summarized:

- As shown in Figure 1, semantic modeling and ontology enable to describe complex system entities such as system lifecycle, their conceptual entities and their virtual models formalizing them. Moreover, the topology among virtual models and their real world counterparts is described as well. In this way, we can create the bridge from real world to virtual world in order to build cognitive twins (Lu, Zheng, et al. 2020).
- Ontology enables to represent the topology across MBSE models in order to represent the interrelationships among different modeling entities. Through this way, dependency, traceability and consistency management can be realized.
- Using semantic triple data structure including subject, object and predicate (ontology models), reasoning can be implemented directly for supporting knowledge management and analysis such as Failure Mode and Effects Analysis (FMEA), etc.
- The ontology may be considered as the middle ware for data integration across different modeling languages and tools (Lu et al. 2021).
- Model formalism, hybrid automata system behaviors, property verification, architecture driven (transforming from one architecture model to others) and code generation are always used during MBSE which is required to be developed based on a basic foundation (Lu, Wang, et al. 2020).

Semantic Model-based Systems Engineering based on KARMA

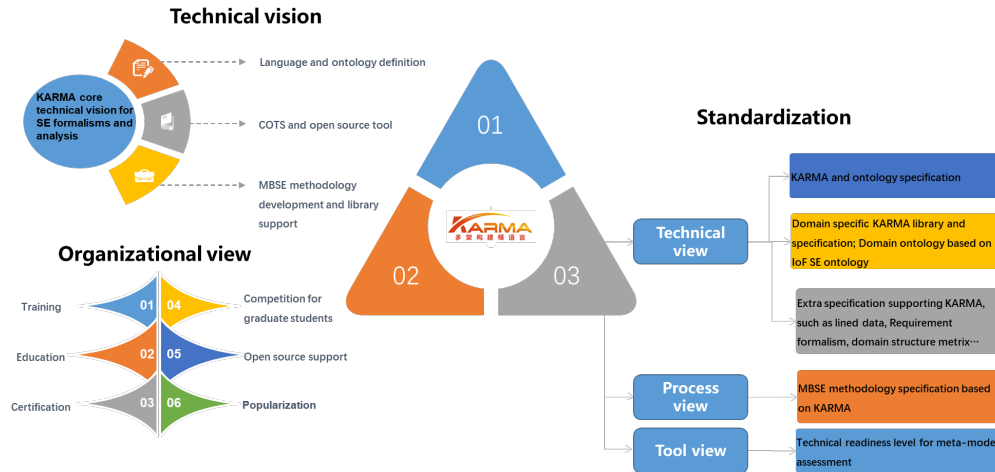


Figure 3 Three cores of KARMA Roadmap

KARMA concepts: KARMA (Kombination of ARchitecture Model specificAtion) LANGUAGE is a semantic specification for a multi-architecture modeling approach. It is developed by EPFL, KTH, Beijing Institute of Technology, Beijing Zhongke Fengchao Ltd, etc enabling to represent meta-model and models based on GOPPRRE meta-meta models, including Graph, Object, Point, Property, Relationship, Role and Extensions (Lu, Wang, et al. 2020). Using the meta-meta models, meta-models of different language specification, such as SysML, OPM, etc, are developed to support MBSE model development. Based on KARMA specification, a commercial-off-the-shelf (COTS) **TOOL** *MetaGraph 2.0* is developed and an open source modeling **TOOL** will be developed in the future. Based on KARMA language and *MetaGraph 2.0*, a MBSE methodology toolbox is developed to support MBSE **METHODOLOGY** design for implementing model configuration and development, such as Requirement-Function-Logic-Physical approach, Mission-Operation-Function-Logic-Physical approach, and etc. In summary, KARMA language provides a semantic specification to construct domain specific meta-models in *MetaGraph 2.0* and the MBSE methodology toolbox is used to develop a working flow based on meta-models in order to implement MBSE methodology.

Three cores of KARMA Roadmap supporting MBSE. As shown in Figure 3, technical vision, organizational view and standardization are three key cores in the KARMA Roadmap:

- The **technical vision** focuses on all the technical aspects related to language and ontology specification, COTS and open source tool development, KARMA library and MBSE methodology.
- The **organization view** focuses on all the organizational support including but not limited to training, education, certification, competition for graduate student, open source support and popularization.
- The **standardization** focus on all the standardized activities including KARMA and ontology specification, domain specific KARMA specification, domain ontology specification, extra support specifications, MBSE methodology specification and technical readiness level specification for meta-model assessment.

In the following, the three cores are introduced separately.

Technical Vision of KARMA Roadmap

Technical vision of KARMA language. As shown in Figure 4, two core mechanisms are proposed to support MBSE research in the technical vision. Based on the proposed GOPPRRE MBSE formalisms (Lu et al. 2021), six key meta-meta models are used to design ontology (W3C Web On-

tology Language , OWL) models based on triple data structure and KARMA specification as an object-oriented and process-oriented programming language. The core features of designed OWL and KARMA specifications are used to support representation of meta-meta models, meta-models and models. Moreover, capabilities of OWL models and KARMA models are shown as follows:

GOPPRRE ontology:

- It supports semantic description of model topology based on a triple data structure.
- It supports consistency management and knowledge management which let stakeholders promote their reusability of the MBSE models and data through semantic reasoning.
- The ontology models enable to be developed as middle ware which support model integration across different MBSE tools.

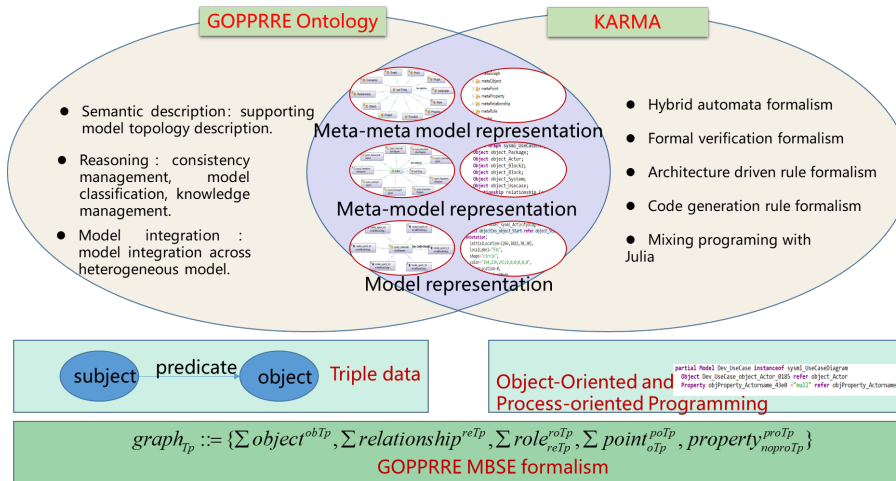


Figure 4 Two cores of KARMA Roadmap

KARMA specification:

- It supports hybrid automata formalism (van Beek et al. 2014) integrated with MBSE formalism. Thus, system behaviors and operational process can be defined within the MBSE models for verification and validation.
- Through defining satisfiability modulo theories (SMT) formalism, KARMA language enables to support formal verifications based on model properties.
- In order to support model transformation among KARMA models and other models, architecture driven rule and code-generation rules can be defined based on KARMA specification (Guo et al. 2020).
- Based on mixed programming with Julia (Elmqvist, Ab, and Lund n.d.), domain specific system analysis, such as complexity analysis, can be implemented integrated with KARMA models.

KARMA language and IoF SE ontology. In order to support full lifecycle systems engineering activities, KARMA language is used to develop different domain specific meta-model libraries for supporting project and process management, mission and system of systems analysis, system architecture design and system lifecycle abilities management, such as fault tree analysis for safety design. In order to create the “bridge” between requirement and MBSE model, ReqIF requirements (Ebert and Jastram 2012) can be integrated with KARMA models through design structure matrix (DSM) table.

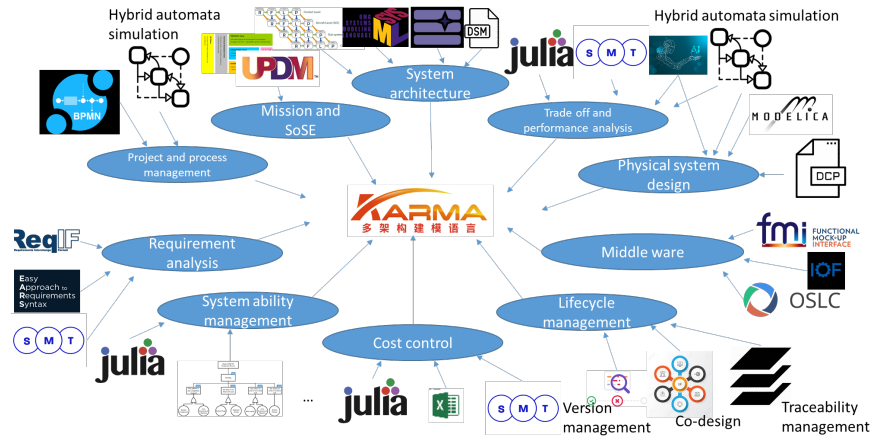


Figure 5 KARMA centric systems engineering techniques

Except for supporting MBSE formalism, systems engineering analysis can be implemented through different KARMA core techniques. Trade-off, performance analysis, cost analysis and control are implemented through mixing programming with Julia, SMT analysis and hybrid automata simulation. Through code-generation, KARMA models are generated into Modelica (Pop and Herzog 2009) and distributed co-simulation protocol (Modelica 2019) execution code in order to support physical system design. Middle ware, such as FMI (Pop and Herzog 2009), IoF ontology (Kulvatuny et al. 2018) and open service lifecycle collaboration specification (OASISOpenProject 2018), are used to develop related adaptors to realize transformations them from and to KARMA models. Finally, based on KARMA models, version management, co-design and traceability management of MBSE models can be realized.

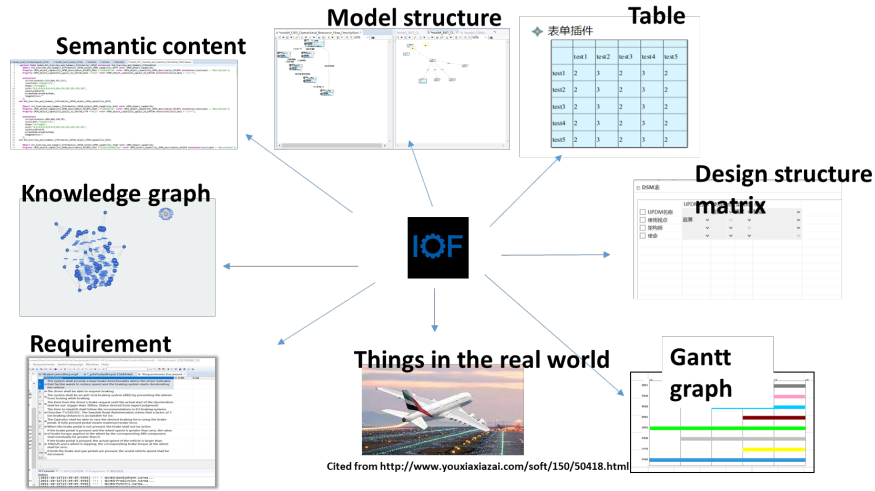


Figure 6 IoF ontology and its representations

The KARMA models and IoF SE ontology can be transformed to each other in *MetaGraph 2.0*². As shown in Figure 6, the generated ontology models from *MetaGraph 2.0* can represent the semantic contents of KARMA language as MBSE model structures. Moreover, table, design structure matrix, requirement items based on ReqIF (Marko et al. 2015) and Gantt graph are also described by the ontology model. Currently, KARMA models are allowed to generate GOPPRRE ontology based on

² An MBSE tool developed based on KARMA, <http://www.zkhoneycomb.com/>. KARMA models and IoF MBSE ontology for 15288 (Zhang et al. 2021) : <https://www.youtube.com/watch?v=3yoSSejbgGI>. KARMA models for architecture design of autonomous driving system: <https://www.youtube.com/watch?v=FlccxJBdtwo&t=58s>.

the BFO top level ontology models. Such models can be redefined to add more entities to represent ontology concepts of real world. In order to realize the unified description of real world and virtual world, the ontology models can be extended to represent real-world entities under the basic formal ontology (Arp et al. 2016). Finally, all the real world and virtual things are represented by the knowledge graph.

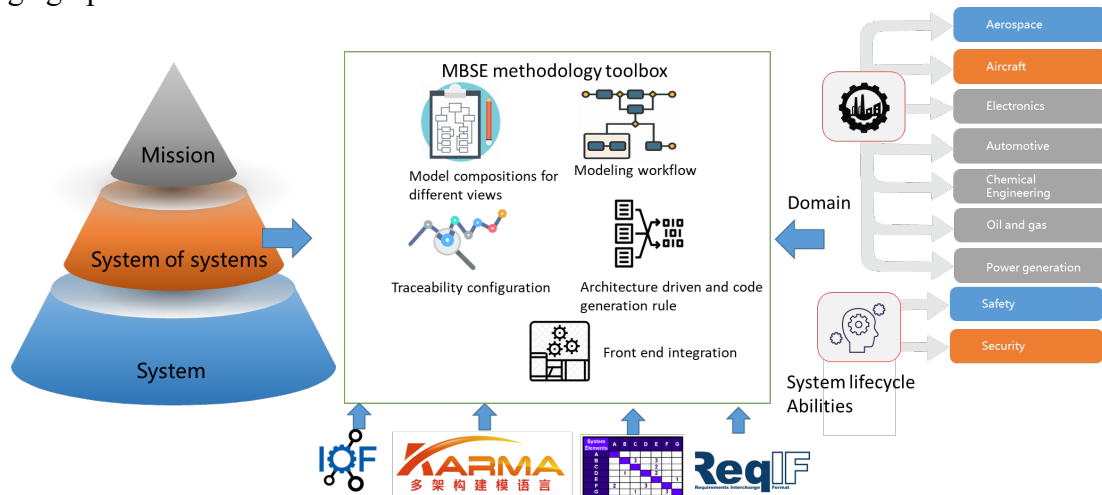


Figure 7 MBSE methodology toolbox for industry practice

Domain specific MBSE methodology for industry. As shown in Figure 7, MBSE methodology toolbox is used to support MBSE from the perspectives of mission, system of systems and system. Moreover, several domains including aerospace, aircraft, electronics, automotive and chemistry engineering, etc, will be considered as the principle domains for practice. Safety and security will be two core focuses from the perspective of system lifecycle “abilities”. In the MBSE toolbox, five steps are implemented based on the basic data format including IoF ontology, KARMA language, DSM table, and ReqIF:

- Develop meta-models for different domain and “system lifecycle abilities” in order to represent the architectural views of the target systems.
- Develop workflow to organize such architecture models based on the developed meta-models.
- Identify the traceability configuration from different models and meta-models, such as one model composition is mapping to others.
- Define the model transformation including architecture driven and code-generation to realize the design automation during MBSE modeling.
- Integration of the previous four steps into front-end based on a standardized plugin development specification.

Simulation, analysis and trade-off. As shown in Figure 8, KARMA technical vision enables model verification and simulation, model analysis and trade-off for the complex system development:

- Model verification and simulation includes static verification and dynamic verification. These are implemented by formal verification based on SMT and hybrid automata simulation. Static verification is mainly used for mission metric validation, system capability validation and system property verification. Dynamic verification is mainly used analysis the system dynamics in SoS, system behaviors and process dynamics.

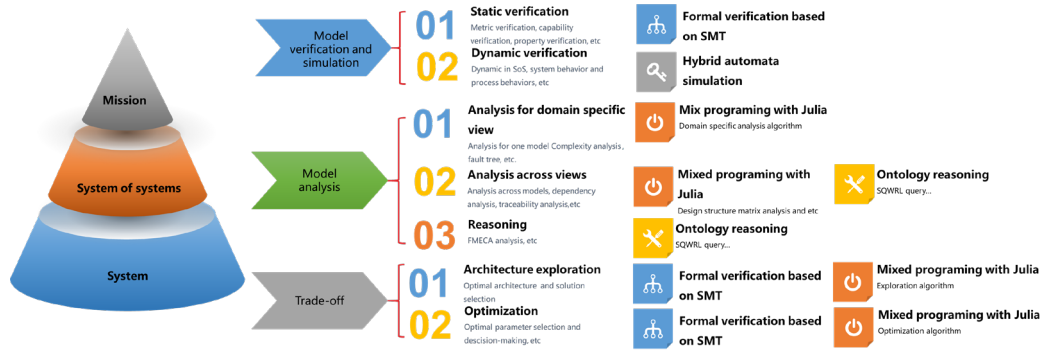


Figure 8 Analysis, simulation and trade-off

- Model analysis includes analysis for the domain-specific view, analysis across views and reasoning. The analysis for the domain specific view refers to analysis for a model, such as complexity analysis and fault tree analysis. This is implemented by mixed programming with Julia. The analysis across views refers to an analysis process for multiple models, such as dependency analysis and traceability analysis. This is implemented based on mixed programming with Julia and ontology reasoning. Reasoning refers to analysis based on ontology reasoning, including Failure Mode, Effects and Criticality Analysis (FMECA) analysis, etc.
- Trade-off is implemented to support architecture exploration and optimization based on SMT analysis and mixed programming with Julia. It can be used for identifying the optimal solutions and selecting the optimal parameters.

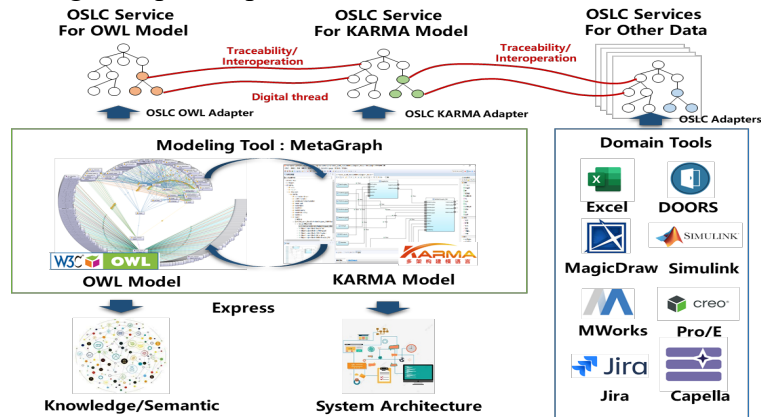


Figure 9 Ontology, KARMA and OSLC service

Tool integration for KARMA. As shown in Figure 9, KARMA language enables to support tool integration by using Open Service for Lifecycle Collaboration (OSLC), for improving the interoperability of heterogeneous models and KARMA models across complex system development:

- In the semantic MBSE approach based on KARMA, KARMA models mainly represent specific information of real world from the architectural perspective, such as the requirement, system architecture, verification and validation. Ontology models generated from KARMA models mainly represent the really world as knowledge from ontology and knowledge perspective.
- By developing specific tool adapters based on OSLC specification and KARMA language, related models are encapsulated into web services, which have complete semantic descriptions of model and data. Besides, OSLC adapters generate OSLC entities across the KARMA tool and other domain specific MBSE tools, which is enabled by linked-data RESTful APIs.
- All the generated OSLC services are used to construct digital thread which can support knowledge management and traceability management of MBSE models for the entire lifecycle.

Organizational views of KARMA Roadmap



Figure 10 Organizational views for KARMA and IoF community

The organizational view of KARMA roadmap aims to provide an organizational support for developing and implementing KARMA core technical vision. As shown in Figure 10, the organizational views include 6 aspects:

- Promoting Systems Engineering Work Station (PSEWS) affiliated with Systems Engineering Society of China (SESC), BIT Systems Engineering International Laboratory Initiatives and EPFL ICT4SM lab take the responsibility of educations on systems engineering, MBSE and ontology engineering and provide training by on-line and off-line lectures.
- Currently, the KARMA team is planning to collaborate with the China MBSE alliance who hold an annual MBSE competition. Then the KARMA tool will be used in the MBSE competition in order to support popularization of KARMA language.
- The IoF systems engineering working group, EPFL ICT4SM lab, BIT Systems Engineering International Laboratory Initiatives, Shanghai Jiaotong University System simulation, optimization, and fault diagnosis lab, KTH mechatronics and University of Oslo SIRIUS group are planning to develop an open-source and web-based MBSE platform based on KARMA and IoF SE and MBSE ontology.
- PSEWS launched a variety of activities. including systems engineering and MBSE webinars, overseas talents systems engineering forum, and systems engineering tour in universities to support MBSE popularization. Moreover, the first version of MBSE white book in Chinese was published last year (Lu Jinzhi 2020).

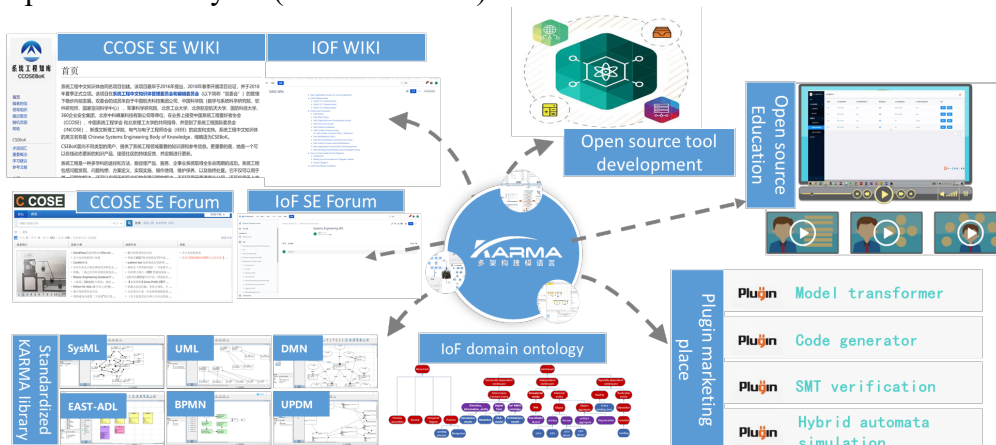


Figure 11 Open source support for KARMA language and IoF SE ontology

As shown in Figure 11, the Promoting Systems Engineering Work Station at Systems Engineering Society of China (CCOSE) and IoF provide wiki and forum for MBSE stakeholders to review the existing MBSE, KARMA and IoF ontology. Moreover, one open source modeling tool based on IoF

ontology and KARMA language is planned to develop by several Chinese and European universities. Open source education resource, such as videos, tutorials and guidance will be published online. Furthermore, IoF ontology and KARMA library will be realized on the web. Finally, a KARMA centric plugin marketing place will be provided for KARMA developers sharing their own knowledge.

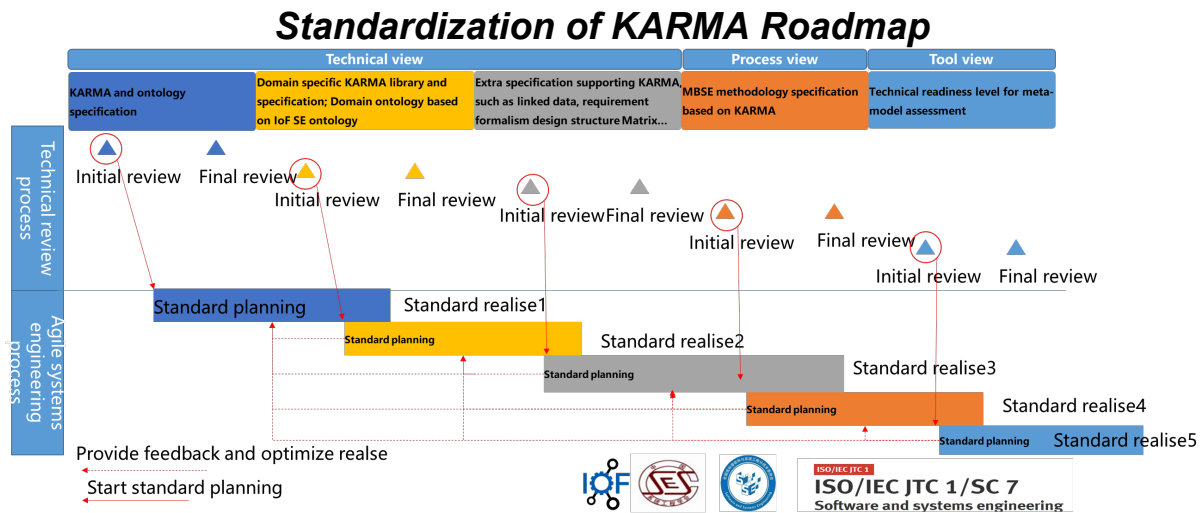


Figure 12 Standardization process for KARMA

Standardizations based on KARMA language and IoF SE and MBSE ontology. As shown in Figure 12, standards from technical view, process view and tool views are developed in a standardization lifecycle led by China National Information Technology Standardization Technical Committee (software and systems engineering) TC28-SC7. First, KARMA and IoF ontology specifications are planned to be developed as the foundation to support the following standard development. Then domain specific KARMA library (domain specific meta-model specification) and domain ontology based on IoF SE ontology specification are planned to be developed as the domain specific application standards for different industries. At the same time, extra specifications supporting KARMA, such as linked data, requirement formalism and design structure matrix, will be developed as China national standards. Such standards are the basic to support technical vision of KARMA tool-chains.

After technical standards are developed, standards for process definition will be developed. MBSE methodology and workflow to implement KARMA models will be concerned during developing such standards. Finally, standards on tool views will be developed. Technical readiness level of meta-models, MBSE tool capabilities and requirements will be concerned in this phase.

Agile systems engineering supporting standardization. Because such standards are strongly dependent to each other, an agile systems engineering process will be implemented when developing such standards. During the technical review process, when initial review of each standard is done, standard planning will be launched until these standards are realized. Moreover, the following standards will be used to provide feedback for the previous standards in order to iterate the standardization planning process for optimizing them.

Discussion

Prospect. In several EU and CN projects, such as H2020 Onto-commons (<https://ontocommons.eu/>), H2020 Qu4lity (<https://qu4lity-project.eu/>), Innosuisse (Project 35258.1) project, China National Key Research and Development Plan (2020YFB1708100), semantic MBSE based on KARMA and IoF ontology is used for aerospace and aircraft industry. Further practices will be done in the future.

Proof-Of-Concept. In H2020 Qu4lity project, sMBSE is implemented for aircraft assembling process modeling (Xiaodu et al. 2022); in Swiss Innosuisse (Project 35258.1 IP-SBM), sMBSE is used

to support mission analysis of ferry analysis by using multi-agent simulation (Meierhofer et al. 2021). The sMBSE is used to support production scheduling based on KARMA language and SMT (Chen et al. 2022). In China National Key Research and Development Plan project (2020YFB1708100), aircraft onboard-maintenance-systems and aircraft development lifecycle are modeled based on KARMA (SysML and BPMN) meta-models (Wang et al. 2021).

Gap and limitation. Since the MBSE practices using KARMA from 2018, we obtained several lessons learned from the tool development, specification definition and industrial practices from implementing the technical aspects. First, from the industrial perspective, modeling languages and tools are less important compared with domain specific views represented by models, particularly that such domain specific modeling tools have more powerful plugins to tell engineers what they need to do. Second, natural language discrepancy is a big challenge when un-English countries make use of MBSE tools. Misunderstandings on language specification, tool operation guidance, and MBSE methodology always lead to failure on MBSE practices. Finally, when developing standards for semantic MBSE, several topics are required to be considered from technical aspects, including 1) language specification, 2) domain specific modeling library specification, 3) ontology specification, 4) MBSE formalisms, 5) MBSE methodology. The consistency between such standardization activities is a big challenge to the standardization activities and organizations. Agile systems engineering is required to manage the such standardization activities.

Future work and plan. The proposed research and practice roadmap will be implemented including training activities for basic systems engineering, MBSE and KARMA language. Moreover, standardization activities will be implemented to develop national and ISO standards for KARMA and MBSE methodology. A web-based open source platform is planned to be developed for the future research roadmap of KARMA and IoF SE ontology. In order to construct the KARMA community, a future work will be developed to integrate more specifications, such as SpecIF (Dungern 2019). Model transformers will be developed in order to transform the current SysML models in different modeling tools to KARMA and GOPPRE ontology. A detailed roadmap to define KARMA standards will be decided in order to realize national industrial standard development.

Conclusion

In this paper, we first propose an overview of ontology, semantic models, modeling language, systems engineering, model-based systems engineering. Then from the literature review, we find semantic modeling is expected to support data interoperability when integrating MBSE languages, tools and methodologies. We introduce a two-core mechanism including KARMA language and IoF SE ontology for full lifecycle system engineering and MBSE formalism. Then a research and practice roadmap is introduced from technical vision, organizational support and standardized activities.

Acknowledgement

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References

- Arp, Robert, Barry Smith, and Andrew D. Spear. 2016. *Building Ontologies with Basic Formal Ontology*. Vol. 91. edited by Intergovernmental Panel on Climate Change. Cambridge: The MIT Press.
- van Beek, D. A., W. J. Fokkink, D. Hendriks, A. Hofkamp, J. Markovski, J. M. van de Mortel-Fronczak, and M. A. Reniers. 2014. "CIF 3: Model-Based Engineering of Supervisory Controllers." Pp. 575–80 in *Lecture Notes in Computer Science (including subseries Lecture*

Notes in Artificial Intelligence and Lecture Notes in Bioinformatics).

- van Beek, D. A., K. L. Man, M. A. Reniers, J. E. Rooda, and R. R. H. Schiffelers. 2006. "Syntax and Consistent Equation Semantics of Hybrid Chi." *The Journal of Logic and Algebraic Programming* 68(1–2):129–210.
- Blekhman, Alex, Juan P. Wachs, and Dov Dori. 2015. "Model-Based System Specification with Tesperanto: Readable Text from Formal Graphics." *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 45(11):1448–58.
- Chen, Jingqi, Guoxin Wang, Jinzhi Lu, Xiaochen Zheng, and Dimitris Kiritsis. 2022. "Model-Based System Engineering Supporting Production Scheduling Based on Satisfiability modulo Theory." *Journal of Industrial Information Integration* 27:100329.
- Dungern, O. .. 2019. *Model-Integration with SpecIF*.
- Ebert, Christof and Michael Jastram. 2012. "ReqIF: Seamless Requirements Interchange Format between Business Partners." *IEEE Software* 29(5):82–87.
- Elmqvist, Hilding, Mogram Ab, and Sweden Lund. n.d. *Modia-A Domain Specific Extension of Julia for Modeling and Simulation*.
- Friedenthal, Sanford, Alan Moore, and Rick Steiner. 2015. "Model-Based Systems Engineering." *A Practical Guide to SysML* 15–29.
- Fritzson, Peter and Vadim Engelson. 1998. "Modelica — A Unified Object-Oriented Language for System Modeling and Simulation." Pp. 67–90 in *Proceedings of the 12th European Conference on Object-Oriented Programming (ECOOP '98)*. Vol. 1445.
- Geiger, Matthias, Simon Harrer, Jorg Lenhard, Mathias Casar, Andreas Vorndran, and Guido Wirtz. 2015. "BPMN Conformance in Open Source Engines." Pp. 21–30 in *2015 IEEE Symposium on Service-Oriented System Engineering*. IEEE.
- Gray, Jeff and Bernhard Rumpe. 2021. "Reflections on the Standardization of SysML 2." *Software and Systems Modeling* 20(2):287–89.
- Guo, Jiangmin, Guoxin Wang, Jinzhi Lu, Junda Ma, and Martin Törngren. 2020. "General Modeling Language Supporting Model Transformations of MBSE (Part 2)." *INCOSE International Symposium* 30(1):1460–73.
- Kulvatunyoo, Boonserm, Evan Wallace, Dimitris Kiritsis, Barry Smith, and Chris Will. 2018. "The Industrial Ontologies Foundry Proof-of-Concept Project." Pp. 402–9 in *IFIP Advances in Information and Communication Technology*.
- Li, Zihang, Jinzhi Lu, Guoxin Wang, Lei Feng, Didem Gurdur Broo, and Dimitris Kiritsis. 2021. "A Bibliometric Analysis on Model-Based Systems Engineering." Pp. 1–8 in *2021 IEEE International Symposium on Systems Engineering (ISSE)*. IEEE.
- Lu Jinzhi, et al. 2020. 历史视角的"基于模型的系统工程" (第一版).
- Lu, Jinzhi, Junda Ma, Xiaochen Zheng, Guoxin Wang, Han Li, and Dimitris Kiritsis. 2021. "Design Ontology Supporting Model-Based Systems Engineering Formalisms." *IEEE Systems Journal* 14(8):1–12.
- Lu, Jinzhi, Guoxin Wang, Junda Ma, Dimitris Kiritsis, Hang Zhang, and Martin Törngren. 2020. "General Modeling Language to Support Model-based Systems Engineering Formalisms (Part 1)." *INCOSE International Symposium* 30(1):323–38.
- Lu, Jinzhi, Xiaochen Zheng, Ali Gharaei, Kostas Kalaboukas, and Dimitris Kiritsis. 2020. "Cognitive Twins for Supporting Decision-Makings of Internet of Things Systems." *Lecture Notes in Mechanical Engineering* 105–15.
- Luisi, James V. 2014. "Information Architecture." Pp. 189–261 in *Pragmatic Enterprise Architecture*. Elsevier.
- Mario Augusto Bunge. 1980. *Treatise on Basic Philosophy - Ontology II: A World of Systems*. Vol. 4. Springer Science & Business Media.

- Marko, Nadja, Andrea Leitner, Beate Herbst, and Alfred Wallner. 2015. "Combining Xtext and OSLC for Integrated Model-Based Requirements Engineering." *Proceedings - 41st Euromicro Conference on Software Engineering and Advanced Applications, SEAA 2015* 143–50.
- MATLAB. n.d. "MATLAB Language Specification." Retrieved (<https://www.mathworks.com/products/matlab.html>).
- Meierhofer, Jürg, Lukas Schweiger, Jinzhi Lu, Simon Züst, Shaun West, Oliver Stoll, and Dimitris Kiritsis. 2021. "Digital Twin-Enabled Decision Support Services in Industrial Ecosystems." *Applied Sciences* 11(23):11418.
- Mengru Dong, Lu Jinzhi, Guoxin Wang, Xiaochen Zheng, Dimitris Kiritsis. 2022. "Model-Based Systems Engineering Analysis Based on Word Cloud Visualization." P. 8 in *preprint*.
- Micouin, Patrice. 2014. *Model Based Systems Engineering : Fundamentals and Methods*. John Wiley & Sons.
- Modelica, Association. 2019. *Distributed Co-Simulation Protocol (DCP)*.
- OASISOpenProject. 2018. *Open Services for Lifecycle Collaboration Core Specification Version 3.0*.
- Pop, Adrian and Erik Herzog. 2009. "Tutorial on SysML , Modelica , Eclipse and ModelicaML." *Systems Engineering*.
- Roques, Pascal, Pascal Roques, Tool European, and Pascal Roques. 2016. "MBSE with the ARCADIA Method and the Capella Tool To Cite This Version : MBSE with the ARCADIA Method and the Capella Tool Keywords : Abstract :". P. 11 in *8Th European Congress on Embedded Real Time Software and Systems (ERTS 2016)*. Toulouse.
- Sztipanovits, Janos, Ted Bapty, Sandeep Neema, Larry Howard, and Ethan Jackson. 2014. "OpenMETA: A Model- and Component-Based Design Tool Chain for Cyber-Physical Systems." *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 8415 LNCS:235–48.
- Wang, Hao, Shaofan Zhu, Jian Tang, Jinzhi Lu, Junpeng Wu, and Dimitris Kiritsis. 2021. "Model-Based Systems Engineering Supporting Cost Analysis of Aircraft Development Process." Pp. 1–3 in *2021 IEEE International Symposium on Systems Engineering (ISSE)*. IEEE.
- Zhang, Yangyang, Zengzhi Liu, Qiang Han, and Wenyuan Zhang. 2021. "Development of System Life Cycle Processes Standardization and Future Evolution Analysis †." Pp. 240–46 in *2021 8th International Conference on Dependable Systems and Their Applications (DSA)*. IEEE.

Biography

Jinzhi Lu	CSEP, is a research scientist at EPFL. He received his PhD at KTH Royal Institute of Technology, Mechatronics Division in 2019. His research interest is MBSE tool-chain design and MBSE enterprise transitioning.
Guoxin Wang	Professor and director of Industrial Engineering Institute at Beijing Institute of Technology. His research interests include reconfigurable manufacturing systems and knowledge-based Engineering.
Yan Yan	Professor at the Institute of Industrial and Intelligent System Engineering, School of Mechanical Engineering, Beijing Institute of Technology, majoring in mechanical engineering.
David Cameron	Coordinator of the SIRIUS Centre for Scalable Data Access at the University of Oslo. He is a chemical engineer with a special interest in knowledge representation, simulation and optimization of chemical process systems.
Feng Lei	Lei Feng (Member, IEEE) joined the Mechatronics and Embedded Control System Division, the KTH Royal Institute of Technology, Stockholm, Sweden, where he is currently an Associate Professor.

Yangyang Zhang	Professor Senior Engineer in the Software application and service research center, China Electronics Standardization Institute, Beijing, China.
Xijin Tang	Xijin Tang is a full professor in the Academy of Mathematics and Systems Science, Chinese Academy of Sciences. She is the secretary general of Systems Engineering Society of China since November of 2018.
Jian Wang	Professor in the DSP Laboratory, University of Electronic Science and Technology of China, Chengdu, China. His current research focuses on system model, design and security.
Junjie Tang	A senior engineer of China Aerospace Science and Technology Corporation. His research focus is on formal modeling and simulation. Dr. Tang holds a PhD in mechanical engineering from Huazhong University of Science and Technology.
Jian Tang	Director of Advanced Avionics Research Department of BATRI, COMAC. His current research interests include embedded complex computing system, MBSE application in avionics domain.
Huisheng Zhang	Professor at the Turbomachinery Institute, School of Mechanical Engineering in Shanghai Jiao Tong University.
Michel Reniers	Michel Reniers (IEEE, senior member) is associate professor at Eindhoven University of Technology. His main research interest is in model-based engineering, discrete-event systems, supervisory control (synthesis) and cyber-physical systems engineering.
Martin Törngren	Martin Törngren (Senior Member, IEEE) was a Visiting Scholar with the UC Berkeley and the Stevens Institute of Technology. Since 2002, he has been a Professor of embedded control systems with the Mechatronics Division, Department of Machine Design, KTH Royal Institute of Technology, Sweden.
Junda Ma	Ph.D. student at the Institute of Industrial and Intelligent System Engineering, School of Mechanical Engineering, Beijing Institute of Technology, majoring in mechanical engineering.
Yves Keraron	Founder of ISADEUS (Innovation Strategy And Digital Engineering Using Semantics) has always been involved in innovation for industry, first in the design and development of nuclear facilities, then in the technical digital information and documentation systems.
Shouxuan Wu	Ph.D. student at the Institute of Industrial and Intelligent System Engineering, School of Mechanical Engineering, Beijing Institute of Technology, majoring in mechanical engineering.
Dimitris Kiritsis	A faculty member at the Institute of Mechanical Engineering of the School of Engineering of EPFL, Switzerland, where he is leading a research group on ICT for Sustainable Manufacturing.